Guichon Creek Raven HBV-EC Model

Model summary: data, inputs, structure, calibration and solar scenario

m.ah a temEEwuh Solar Project · Swiftwater Consulting scope SWC-1-43-01-P-001 · generated 2026-11-12

Purpose

Assess how the m.ah a temEEwuh Solar Project's land-cover change affects flows in Guichon Creek, BC, about 5 km north of Logan Lake. The deliverable is before/after hydrographs at a model outlet on Guichon Creek immediately below the solar site. A small change is expected, so results are reported as differences between runs, not absolute flows.

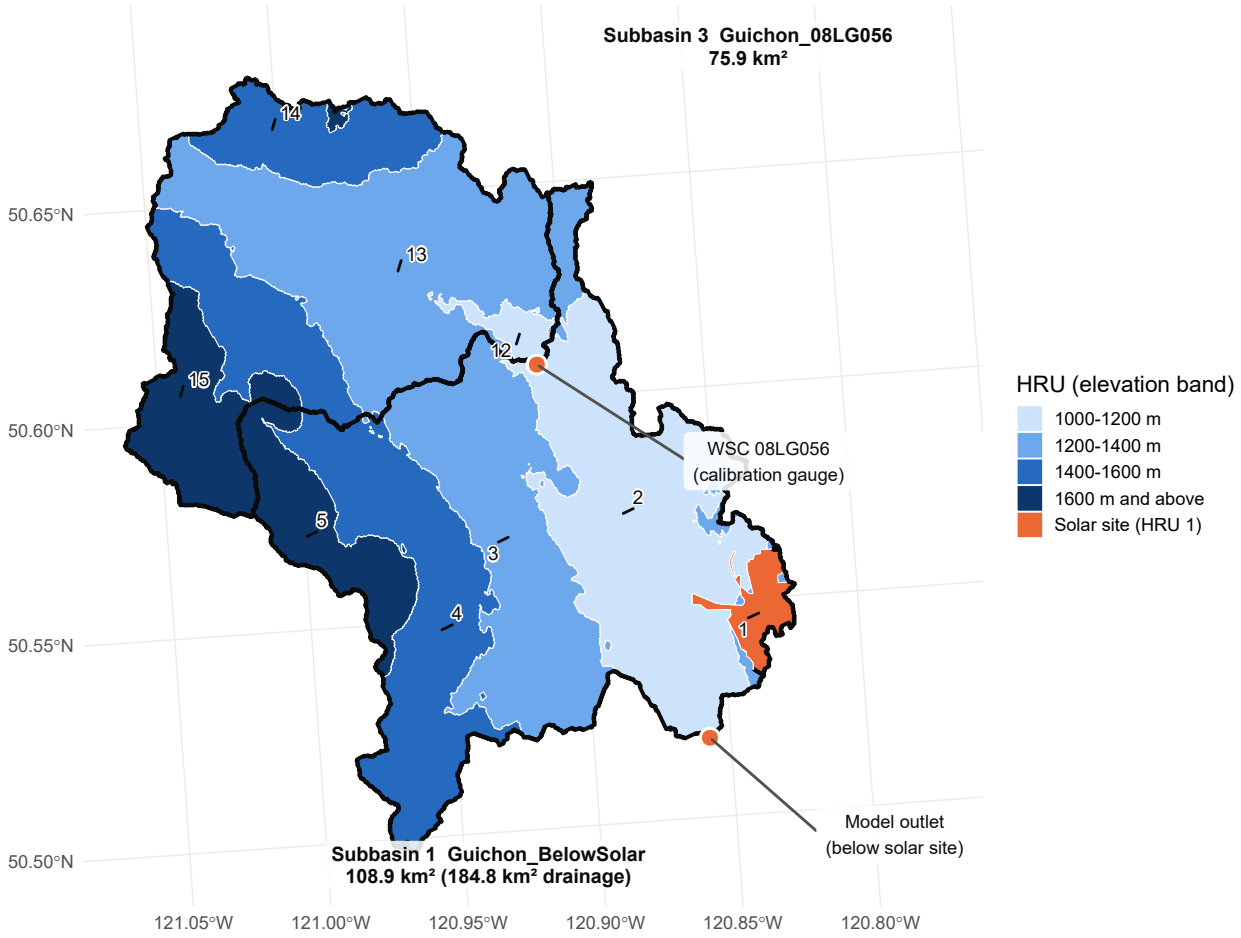
Model at a glance

Item	Description
Model	Raven v4.1 (build 2025-12-08), HBV-EC emulation (manual App. F.2)
Domain	Guichon Creek to the outlet below the solar site: 184.8 km² in 2 subbasins
Spatial units	9 HRUs: 200 m elevation bands, split at WSC 08LG056, plus the solar site (HRU 1, 3.24 km²)
Forcing	Daymet V4 R1 gridded Tmax/Tmin/precip, area-averaged into one virtual gauge per HRU
Time step and period	Daily, 1980-10-01 to 2024-09-30 (WY1981 spin-up)
Calibration	OSTRICH DDS, 1000 trials, 12 parameters, maximize KGE at WSC 08LG056 for WY2005–2024
Fit at 08LG056 (KGE)	0.86 calibration (WY2005–2024), 0.08 validation (WY1982–2004)
Solar scenario	Solar site from 20% forest to fully impervious (IMPERM 1.0): worst-case screening
Result (model outlet)	Mean annual flow +3.3%, mean annual peak +2.4% (WY1982–2024)

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2 Catchment and HRUs



Map geometry: project geodatabase (hrus_v3_08LG056, cleaned 2026-11-12). Subbasin 3 is the catchment of WSC 08LG056 cut out of subbasin 1 with the CLRH hydrofabric; HRUs 2–5 are the parts of each elevation band below the gauge and HRUs 12–15 the parts above it. HRU IDs follow the GIS layers. The model outlet is the downstream end of subbasin 1 (50.5184 N, 120.8576 W).

Subbasins

ID	Name	Downstream	Area (km²)	Drainage area (km²)	Mean elev. (m)	TIME_CONC / TIME_TO_PEAK (d)	Gauge
1	Guichon_BelowSolar	outlet	108.91	184.81	1,326	2 / 1	none (deliverable outlet)
3	Guichon_08LG056	1	75.90	75.90	1,429	1.3 / 0.65	WSC 08LG056 (calibration)

HRUs (hbvec.rvh)

HRU	Subbasin	Band	Area (km²)	Elev. (m)	Lat	Lon	Land use / vegetation	Sample points
1	1	Solar site	3.237	1,215	50.5481	-120.8366	MIX20_LU / MIX20_VEG	50
2	1	1000–1200 m	37.193	1,151	50.5686	-120.8767	MIX20_LU / MIX20_VEG	9
3	1	1200–1400 m	31.973	1,292	50.5721	-120.9211	MIX20_LU / MIX20_VEG	8
4	1	1400–1600 m	25.580	1,478	50.5523	-120.9631	MIX20_LU / MIX20_VEG	6
5	1	1600 m +	10.927	1,692	50.5734	-120.9927	MIX20_LU / MIX20_VEG	175
12	3	1000–1200 m	2.135	1,186	50.6167	-120.9194	MIX20_LU / MIX20_VEG	34
13	3	1200–1400 m	39.078	1,308	50.6355	-120.9651	MIX20_LU / MIX20_VEG	10
14	3	1400–1600 m	22.488	1,494	50.6420	-121.0005	MIX20_LU / MIX20_VEG	6
15	3	1600 m +	12.203	1,743	50.6030	-121.0333	MIX20_LU / MIX20_VEG	197

3 Data sources

Dataset	Source	Used
Elevation	NRCan MRDEM 30 m DTM (data/gis/MRDEM30_dtm.tif)	HRU elevation, area, map
Subbasins and HRUs	GIS delineation from the DEM in ArcGIS Pro (hrus_final.shp, subbasins_final.shp); attributes by hru_attributes_arcpy.py	Model geometry (.rvh)
08LG056 catchment	Canadian Lake-River Hydrofabric (CLRH) finalcat_info v1-1, catchments draining to Obs_NM = 08LG056 (split_08LG056_subbasin.R)	Subbasin 3 / HRUs 12-15
Streamflow	WSC HYDAT 08LG056 daily mean discharge via ECCC GeoMet API, retrieved 2026-11-12; 1982-01-01 to 2024-12-31 (get_obs_08LG056.py)	Calibration target
Meteorology	Daymet V4 R1 1 km daily Tmax, Tmin, precip (Thornton et al. 2022, doi:10.3334/ORNLDAAC/2129), single-pixel API, 1980-2024; 495 sample points in the modelled HRUs (get_daymet.R)	hbvec.rvt (one gauge per HRU)
Soil porosity	BC Soil Information Finder Tool (SIFT) texture mix with Rawls et al. (1982) porosities, less 23% coarse fragments: POROSITY 0.36	hbvec.rvp
Forest cover, land use	Assumed 20% forest cover for all HRUs (judgment, 2026-11-12); interception rates are Raven defaults	hbvec.rvp
Solar footprint	Project footprint polygon clipped to the watershed (HRU 1)	Solar scenario

Raw downloads are kept unedited in data/raw/; the scripts in data/scripts/ rebuild every model input (see data/README.md). The Daymet point downloads (data/raw/daymet/) are not in git.

4 Model structure

HBV-EC emulation from the Raven v4.1 manual (Appendix F.2). Each HRU has a snowpack, a canopy, a soil moisture box (TOPSOIL, SOIL[0]) and two conceptual groundwater stores: the fast (upper zone, SOIL[1]) and slow (lower zone, SOIL[2]) reservoirs. Runoff from each HRU is routed to its subbasin outlet with a triangular unit hydrograph; there is no channel routing between subbasins (ROUTE_NONE), so subbasin 3's outflow is added to subbasin 1's.
Key options: :Method ORDERED_SERIES; :Interpolation INTERP_NEAREST_NEIGHBOR; :Evaporation PET_FROMMONTHLY (monthly Hargreaves normals per gauge); :OroTempCorrect / :OroPrecipCorrect / :OroPETCorrect OROCORR_HBV (zero lapse, because each gauge sits at its HRU's elevation); :PrecipIceptFract PRECIP_ICEPT_USER; :SoilModel SOIL_MULTILAYER 3.

Hydrologic processes (hbvec.rvi)

Process	Algorithm
Snow refreeze	FREEZE_DEGREE_DAY
Precipitation partitioning	PRECIP_RAVEN, RAINSNOW_HBV
Canopy evaporation / sublimation	CANEVP_ALL (intercepted water all evaporates)
Snow balance	SNOBAL_SIMPLE_MELT, POTMELT_HBV
Snow liquid overflow	OVERFLOW_RAVEN
Infiltration	INF_HBV (IMPERM bypasses soil)
Flush to upper zone	RAVEN_DEFAULT
Soil evaporation	SOILEVAP_HBV
Capillary rise	CRISE_HBV
Percolation	PERC_CONSTANT
Fast baseflow	BASE_POWER_LAW (FAST_RESERVOIR = SOIL[1])
Slow baseflow	BASE_LINEAR (SLOW_RESERVOIR = SOIL[2])
Within-subbasin routing	ROUTE_TRI_CONVOLUTION
Channel routing	ROUTE_NONE

Scenario method

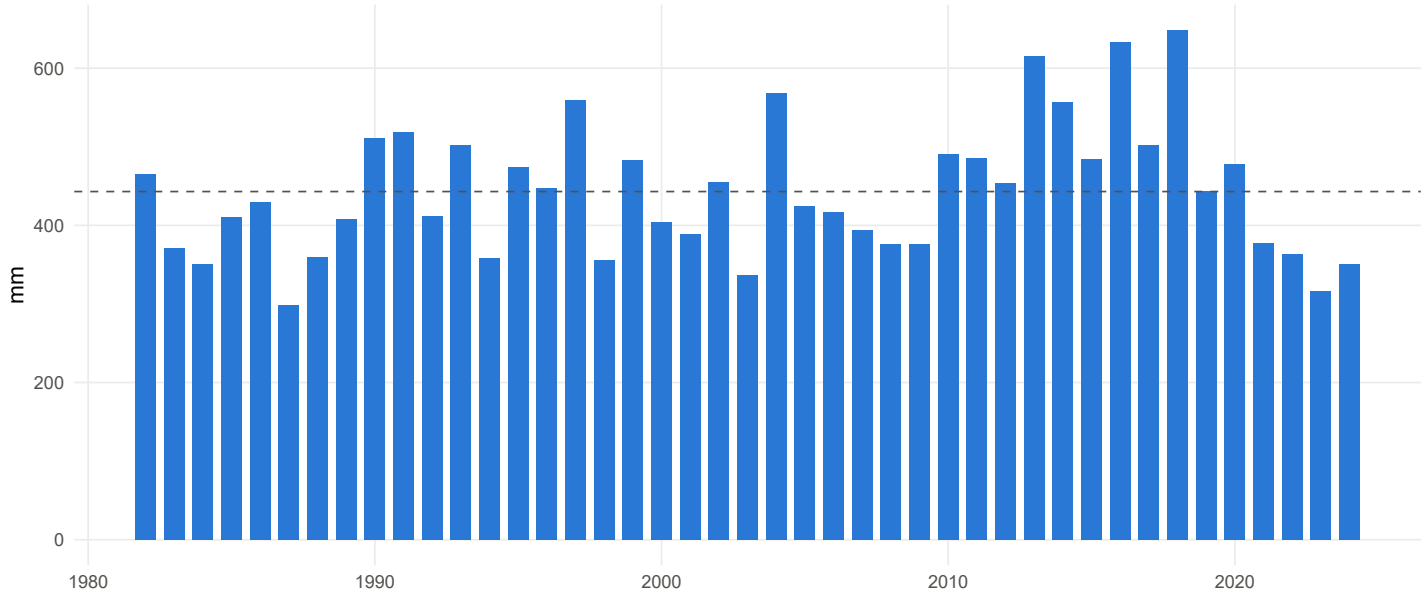
hbvec.rvp is the baseline. hbvec_solar.rvp includes it (:RedirectToFile hbvec.rvp) and adds :LandUseChange SOLAR_SITE SOLAR and :VegetationChange SOLAR_SITE SOLAR_VEG dated 1900-01-01, which Raven applies at t = 0. The two runs therefore share every parameter and forcing and differ only in the solar HRU's land cover.

5 Forcing: Daymet virtual gauges

- Daymet V4 R1 (1 km daily grid) was sampled on a 2 km grid in each HRU (250 m where an HRU would get fewer than 5 points) through the open single-pixel API, because the NetCDF server needs a NASA Earthdata login. 495 points cover the modelled HRUs.
- Each HRU's daily Tmax, Tmin and precipitation is the mean of its points; the grid is equal-area, so this is the area-weighted HRU mean, as Raven's :GriddedForcing + :GridWeights would compute.
- hbvec.rvt holds one gauge per HRU (Daymet_HRUnn) at that HRU's own latitude, longitude and elevation. Nearest-neighbour interpolation therefore gives each HRU its own gauge, and OROCORR_HBV adds no lapse.
- All gauges share :RainCorrection 0.805 and :SnowCorrection 0.902 (calibrated). Monthly temperature and Hargreaves PET normals are computed per HRU over WY1981–2024.
- Daymet omits Dec 31 in leap years; it is filled with Dec 30's temperatures and zero precipitation.

Model-area precipitation (Daymet, HRU area-weighted), by water year

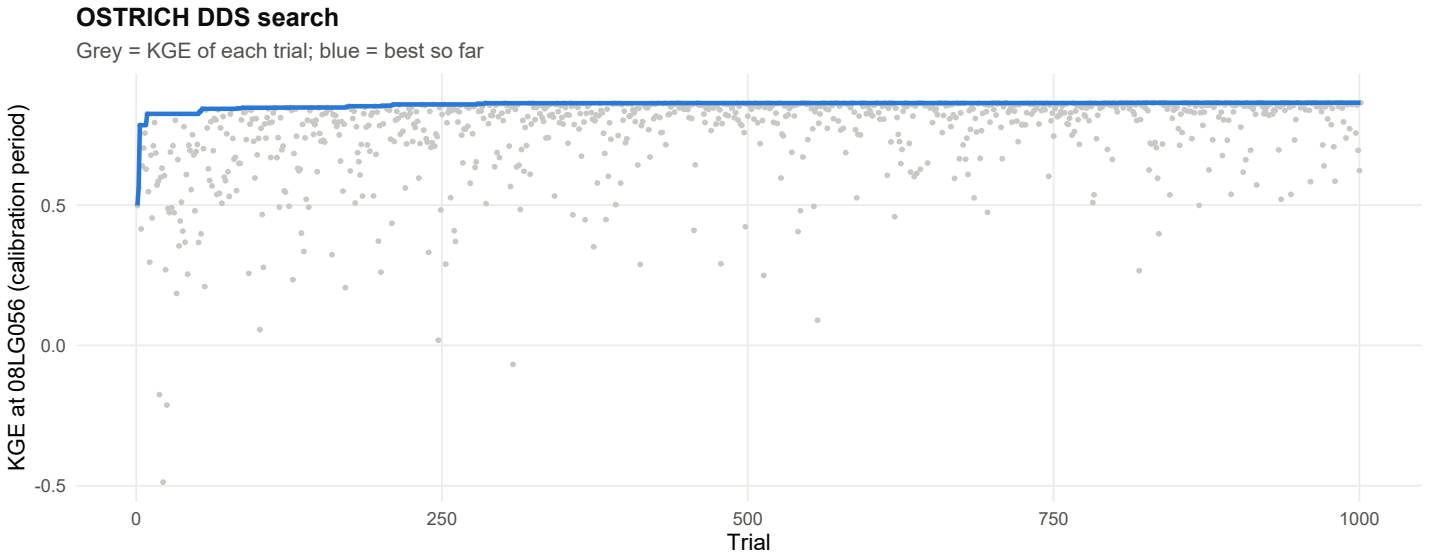
Before :RainCorrection/:SnowCorrection. Mean 443 mm/yr (dashed); mean temperature 3.1 °C.



6 Calibration

- Target: WSC 08LG056, Guichon Creek above Tunkwa Lake Diversion (76.06 km², natural flow, RHBN reference station), the outlet of subbasin 3.
- Objective: maximize Kling-Gupta efficiency (KGE) of daily flow over the CALIBRATION period, WY2005–2024 (present-day conditions, which the project acts on). VALIDATION is WY1982–2004. Observed flow at 08LG056 roughly doubled after about 2004 while precipitation did not (pine beetle, mine operations suspected but unverified), so the two periods are different regimes.
- Method: OSTRICH v17.12.19, Dynamically Dimensioned Search (DDS), 1000 trials, perturbation 0.20, random seed 20261112, started from the values of an earlier calibration. Setup and checks in calib/ (prepare.py, ostIn.txt); results in calib/results_2026-11-12_daymet/.
- Calibrated: 12 parameters (table below). MIN_MELT_FACTOR is tied as a fraction of MELT_FACTOR so it cannot exceed it. Solar-site parameters are not calibrated.

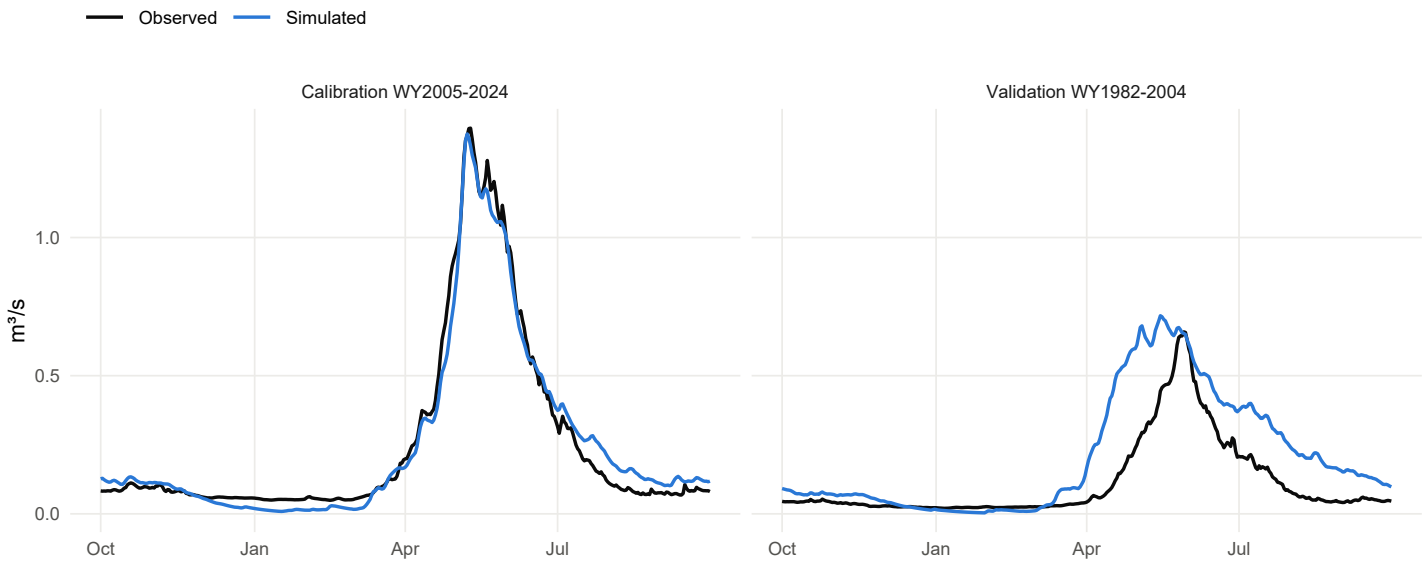
OSTRICH name	Raven parameter	Range	Start	Calibrated (in model)
par_TT	RAINSNOW_TEMP (°C)	-1.000 – 2.500	0.727	0.715
par_CFMAX	MELT_FACTOR (mm/d/K)	2.500 – 7.000	4.384	4.001
par_MINFRAC	MIN_MELT_FACTOR / MELT_FACTOR	0.200 – 1.000	0.209	0.237
par_SFCF	:SnowCorrection	0.800 – 1.500	0.896	0.902
par_RFCF	:RainCorrection	0.800 – 1.300	1.016	0.805 (at bound)
par_THICK	TOPSOIL thickness (m)	0.300 – 2.000	1.456	1.662
par_BETA	HBV_BETA	0.500 – 4.000	2.200	1.017
par_LP	FIELD_CAPACITY (HBV LP)	0.500 – 0.900	0.642	0.798
par_PERC	MAX_PERC_RATE (mm/d)	0.100 – 6.000	1.706	1.213
par_KF	BASEFLOW_COEFF fast (1/d)	0.01 – 0.500	0.0553	0.0446
par_BFN	BASEFLOW_N fast	1.000 – 3.000	1.150	1.173
par_KS	BASEFLOW_COEFF slow (1/d)	0.001 – 0.05	0.0441	0.0498 (at bound)



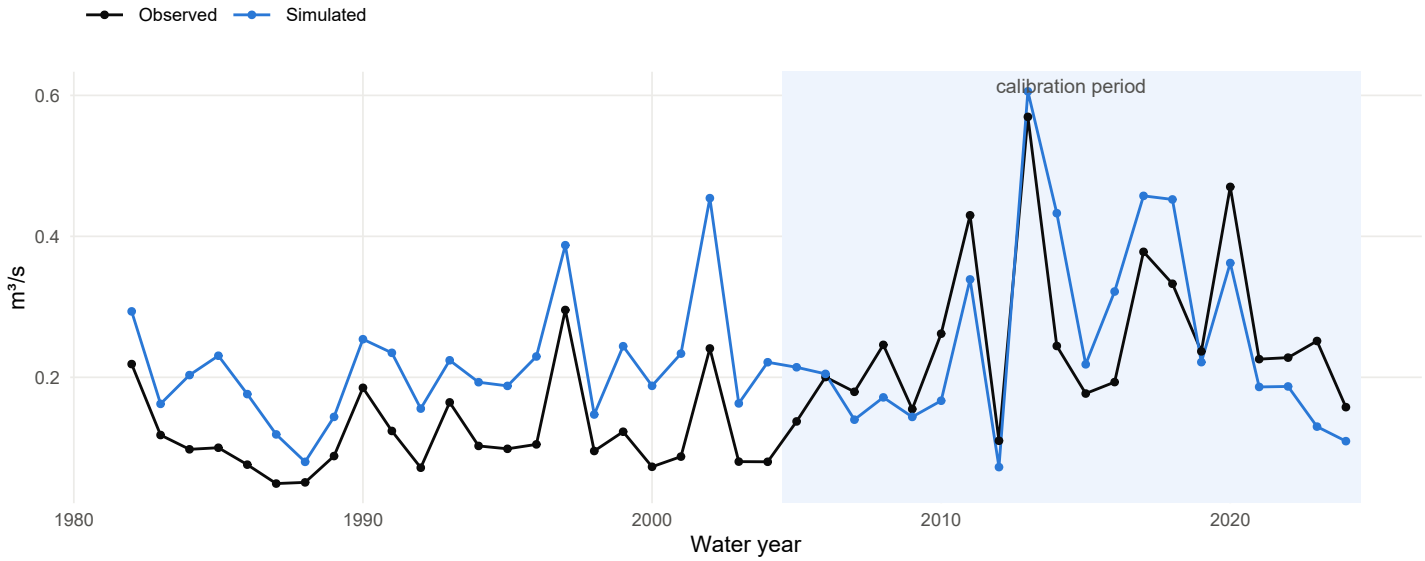
7 Calibrated fit at WSC 08LG056

Period	NSE	KGE	Bias (%)	RMSE (m³/s)	R2
Calibration WY2005–2024	0.73	0.86	-0.5	0.240	0.75
Validation WY1982–2004	0.14	0.08	+81.4	0.198	0.65
All WY1982–2024	0.62	0.67	+28.0	0.218	0.70

Mean daily hydrograph at WSC 08LG056



Annual mean flow at WSC 08LG056 (days with observations)



The calibration reproduces the higher post-2004 flows by letting more water leave the soil (HBV_BETA ≈ 1.0, FIELD_CAPACITY ≈ 0.80), and as a result over-predicts the earlier regime (validation bias about +80%). Diagnostics are from the baseline run with the rounded parameters in hbvec.rvp/.rvt.

8 Final parameters

Calibrated (as written in hbvec.rvp / hbvec.rvt)

Raven parameter	File	Value
RAINSNOW_TEMP (°C)	hbvec.rvp	0.715
MELT_FACTOR (mm/d/K)	hbvec.rvp	4.001
MIN_MELT_FACTOR / MELT_FACTOR	hbvec.rvp	0.237 (MIN_MELT_FACTOR = 0.947 mm/d/K)
:SnowCorrection	hbvec.rvt (all gauges)	0.902
:RainCorrection	hbvec.rvt (all gauges)	0.805 (at bound)
TOPSOIL thickness (m)	hbvec.rvp	1.662
HBV_BETA	hbvec.rvp	1.017
FIELD_CAPACITY (HBV LP)	hbvec.rvp	0.798
MAX_PERC_RATE (mm/d)	hbvec.rvp	1.213
BASEFLOW_COEFF fast (1/d)	hbvec.rvp	0.0446
BASEFLOW_N fast	hbvec.rvp	1.173
BASEFLOW_COEFF slow (1/d)	hbvec.rvp	0.0498 (at bound)

Fixed (not calibrated)

Parameter	Value	Basis
POROSITY	0.36	Soil survey (SIFT)
SAT_WILT	0.0	HBV-EC template
MAX_CAP_RISE_RATE	1.0 mm/d	HBV-EC template
RAINSNOW_DELTA / SNOW_SWI	2.0 °C / 0.05	HBV-EC template
ADIABATIC_LAPSE / PRECIP_LAPSE	6.5 °C/km / 0.0 (no lapse: gauges at HRU elevation)	HBV-EC template; Daymet already elevation-adjusted
HBV_MELT_FOR_CORR / REFREEZE_FACTOR / HBV_MELT_ASP_CORR	0.5 / 2.0 mm/d/K / 0.48	Template / not identifiable from flow
FOREST_COV (all HRUs, baseline)	0.2	Judgment
RAIN_ICEPT_PCT / SNOW_ICEPT_PCT	0.12 / 0.10 (Raven defaults)	Not set explicitly
TIME_CONC / TIME_TO_PEAK	SB1 2.0 / 1.0 d; SB3 1.3 / 0.65 d	Guess (SB3 scaled by sqrt of area)
:AvgAnnualRunoff	75 mm/yr (08LG056 mean)	Reference discharge only

Land use and vegetation classes

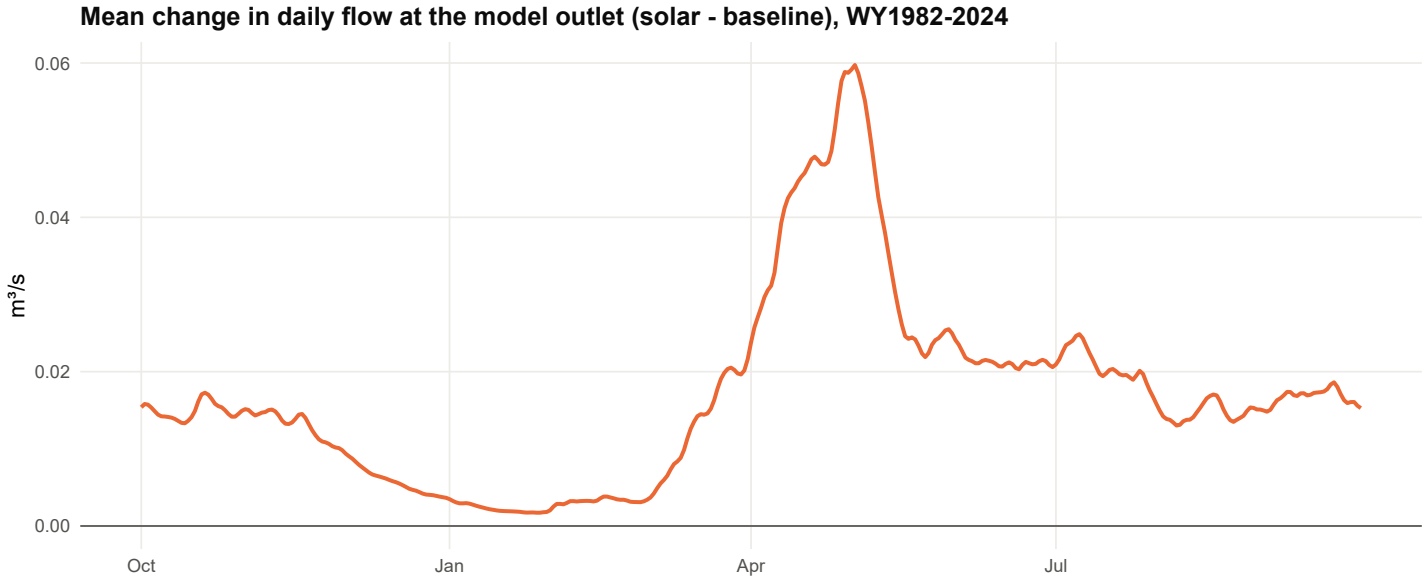
Class	Used for	FOREST_COV	IMPERM	MAX_HT / MAX_LAI
MIX20_LU / MIX20_VEG	All HRUs (baseline), including the solar site	0.2	0.0	5.4 m / 2.0
SOLAR / SOLAR_VEG	Solar site HRU 1 in the solar run only	0.0	1.0	0.5 m / 1.0

MAX_HT and MAX_LAI have no effect in this configuration; forest cover acts only through FOREST_COV (interception losses and the HBV forest melt correction).

9 Solar scenario

- Baseline: the solar site (HRU 1, 3.24 km², in subbasin 1) has 20% forest cover like the rest of the watershed.
- Solar: HRU 1 becomes land use SOLAR (FOREST_COV 0, IMPERM 1.0). All rain and snowmelt on the site bypasses the soil and goes to the fast reservoir. This is a worst-case screening assumption, not an estimate of the real effect.
- The project footprint clipped to the watershed was given as about 2.1 km²; HRU 1 is 3.24 km². If 2.1 km² is right, the modelled change is about 1.5 times too large.

Statistic	Baseline (m³/s)	Solar (m³/s)	Change (m³/s)	Change (%)
Mean annual flow	0.5104	0.5273	+0.0169	+3.3
Mean annual peak daily flow	3.1265	3.2010	+0.0744	+2.4
Mean annual 7-day low flow	0.0048	0.0053	+0.0006	+11.7



Why the solar run is higher every day: removing soil evaporation on the site (roughly 160 mm/yr) adds water to the fast reservoir, which also percolates to the slow reservoir; the slow reservoir drains a fixed fraction daily, so flow rises on every day of the record.
Caveat on low flows: in this HBV structure impervious runoff still recharges the slow reservoir, so the model cannot show the loss of groundwater recharge that a real impervious surface causes. The volume and peak increases are a reasonable worst case; the low-flow increase is a model artifact and its real sign may be negative.

10 Limitations and files

Limitations

- One calibration gauge. 08LG056 drains the higher, wetter headwaters (mostly 1200–1750 m); the solar site (about 1215 m) and the rest of subbasin 1 are lower and drier and are assumed to share its parameters. There is no gauge at the model outlet.
- Regime shift. Calibrated to WY2005–2024; the same parameters over-predict WY1982–2004 by about 80%.
- Two calibrated parameters finished at their bounds (:RainCorrection lower bound 0.80; slow BASEFLOW_COEFF upper bound 0.05).
- Tunkwa Lake diversion just below 08LG056 (historic WSC 08LG014) is not modelled; if it still operates, outlet flows are overstated in both runs.
- Forest cover (20%) and interception rates are assumed, not mapped (BC VRI could replace the assumption).
- No channel routing between subbasins (ROUTE_NONE); daily time step only, so sub-daily storm peaks are not represented.
- The solar scenario is a bound: fully impervious site, no panel or depression storage, no evaporation from impervious surfaces, and recharge not reduced (see section 9).

Run commands (from HBVEC_Model/)

```
Raven.exe hbvec -o output\ -r baseline
Raven.exe hbvec -p hbvec_solar.rvp -o output\ -r solar
Rscript analysis/plot_solar_dygraph.R      (interactive baseline vs solar)
Rscript model_documentation/build_model_summary.R      (this document)
```

Files

File	Contents
hbvec.rvi	Options, periods, processes, outputs
hbvec.rvh	Subbasins, HRUs, HRU groups
hbvec.rvp	Parameters (baseline)
hbvec_solar.rvp	Solar run: includes hbvec.rvp + land-cover change
hbvec.rvt	Daymet gauges and 08LG056 observations (data/model_input/forcing/*.rvt)
hbvec.rvc	Initial conditions (defaults)
data/	Raw data, processed data and build scripts (data/README.md)
calib/	OSTRICH setup and saved results (calib/README.md)
analysis/	Plot scripts
CLAUDE.md	Project conventions, decisions and Raven gotchas